

Electromagnetism & Electrodynamics HW2

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Technion

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1 Delta function identities

1. show the delta function is symmetric

2. Show

$$\delta(\alpha x) = \frac{\delta(x)}{|\alpha|}; \quad \alpha \neq 0 \quad (1)$$

3. show

$$\delta(g(x)) = \sum_{i=1}^N \frac{\delta(x - x_i)}{|g'(x_i)|} \quad (2)$$

4. find

$$\int dx f(x) \delta'(x) \quad (3)$$

5. show that

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{-ikx} \quad (4)$$

1.1 symmetry

Let $f(x)$ be an arbitrary function. Therefore,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \delta(-x) dx &= \left[\begin{array}{l} u = -x \\ du = -dx \end{array} \right] = \int_{\infty}^{-\infty} -f(-u) \delta(u) du \\ &= \int_{-\infty}^{\infty} f(-u) \delta(u) du \\ &= f(-0) = f(0) = \int_{-\infty}^{\infty} f(x) \delta(x) dx \end{aligned}$$

Therefore the delta function is symmetric.

1.2 multiplication

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \delta(\alpha x) dx &= \left[\begin{array}{l} u = \alpha x \\ du = \alpha dx \end{array} \right] = \int_{-sgn(\alpha) \cdot \infty}^{sgn(\alpha) \cdot \infty} \frac{1}{\alpha} f\left(\frac{u}{\alpha}\right) \delta(u) du \\ &= \frac{sgn(\alpha)}{\alpha} f\left(\frac{0}{\alpha}\right) \\ &= \frac{1}{|\alpha|} f(0) \end{aligned}$$

Therefore

$$\delta(\alpha x) = \frac{\delta(x)}{|\alpha|}$$

1.3 delta of function

We'll develop $g(x)$ into a taylor series around x_i :

$$g(x) \approx g(x_i) + g'(x_i) \cdot (x - x_i) = g'(x_i) \cdot (x - x_i)$$

ignoring elements beyond first order. Next we'll look at the following integral:

$$\int_{-\infty}^{\infty} \delta(g(x)) f(x) dx$$

where $f(x)$ is an arbitrary function. It's clear from the definition of $g(x)$ and the delta function that the integrand is non-zero only at the points x_i , therefore we can integrate only around those points. We'll choose a small enough neighborhood for each point so they are isolated:

$$\int_{-\infty}^{\infty} \delta(g(x)) f(x) dx = \sum_{i=1}^N \int_{x_i - \varepsilon_i}^{x_i + \varepsilon_i} \delta(g(x)) f(x) dx$$

Using the delta function definition and the previous subsection we can find:

$$\begin{aligned} \sum_{i=1}^N \int_{x_i - \varepsilon_i}^{x_i + \varepsilon_i} \delta(g(x)) f(x) dx &= \sum_{i=1}^N \int_{x_i - \varepsilon_i}^{x_i + \varepsilon_i} \delta(g'(x_i) \cdot (x - x_i)) f(x) dx \\ &= \left[\begin{array}{l} u = g'(x_i) \cdot (x - x_i) \\ du = g'(x_i) dx \end{array} \right] \\ &= \sum_{i=1}^N \int_{\text{sgn}(g'(x_i)) \cdot g(x_i - \varepsilon_i)}^{\text{sgn}(g'(x_i)) \cdot g(x_i + \varepsilon_i)} \frac{1}{g'(x_i)} \delta(u) f\left(x_i + \frac{u}{g'(x_i)}\right) du \\ &= \sum_{i=1}^N \frac{1}{|g'(x_i)|} \int_{g(x_i - \varepsilon_i)}^{g(x_i + \varepsilon_i)} \delta(u) f\left(x_i + \frac{u}{g'(x_i)}\right) du \\ &= \sum_{i=1}^N \frac{f(x_i)}{|g'(x_i)|} \\ &= \sum_{i=1}^N \int_{-\infty}^{\infty} f(x) \frac{\delta(x - x_i)}{|g'(x_i)|} \end{aligned}$$

Therefore

$$\delta(g(x)) = \sum_{i=1}^N \frac{\delta(x - x_i)}{|g'(x_i)|}$$

1.4 derivative of delta

Integrating by parts we get

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)\delta'(x)dx &= \cancel{f(x)\delta(x)}\Big|_{-\infty}^{\infty} \overset{0}{\rightarrow} - \int_{-\infty}^{\infty} f'(x)\delta(x)dx \\ &= -f'(0)\end{aligned}$$

1.5 fourier transform

If we perform a fourier transform on the delta function we get

$$\mathcal{F}\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x)e^{iky}dy = \frac{1}{\sqrt{2\pi}} e^{iky} \Big|_{y=0} = \frac{1}{\sqrt{2\pi}} \quad (5)$$

therefore if we apply the inverse fourier transform

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{F}\delta(y)e^{iky}dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iky}dy \quad (6)$$

and if we rename the parameters

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx}dk \quad (7)$$

2 Charge Distributions

Write an expression for the volumetric charge distributions for the following distributions in spherical coordinates:

1. A ball of radius R whose center is at the origin and is uniformly charged
2. A spherical shell of radius R placed at the origin, its northern half charged with surface charge σ_N and the southern half charged with σ_S
3. An infinite wire passing through the origin coinciding with $z = y = 0$ and uniformly charged with linear charge density λ

2.1 ball

The volume of a sphere is $V = \frac{4\pi R^3}{3}$ therefore $\rho = \frac{3Q}{4\pi R^3}$ when inside the ball. This means the charge distribution is

$$\rho = \frac{3Q}{4\pi R^3} \Theta(R - r) \quad (8)$$

2.2 shell

A direction calculation

$$\rho = \delta(R - r) \left[\Theta\left(\frac{\pi}{2} - \theta\right) \sigma_N + \Theta\left(\theta - \frac{\pi}{2}\right) \sigma_S \right] \quad (9)$$

2.3 infinite wire

Another direction calculation

$$\rho = \delta\left(\theta - \frac{\pi}{2}\right) [\delta(\phi) + \delta(\phi - \pi)] \lambda(r) \quad (10)$$

$$= \delta\left(\theta - \frac{\pi}{2}\right) [\delta(\phi) + \delta(\phi - \pi)] \frac{\lambda}{r^2} \quad (11)$$

3 Yukawa Potential

The following potential field is given

$$\phi(r) = \frac{1}{r} e^{-Kr}; \quad K > 0 \quad (12)$$

1. Calculate the gradient for $r \neq 0$
2. Calculate the laplacian for $r \neq 0$
3. Use gauss' law to find an expression for the laplacian while includes the origin
4. Show that

$$\iiint d^3x \nabla^2 \phi = 0 \quad (13)$$

3.1 gradient

The gradient in spherical coordinates is defined as

$$\nabla \phi = \partial_r \phi \hat{r} + \frac{1}{r} \partial_\theta \phi \hat{\theta} + \frac{1}{r \sin \theta} \partial_\varphi \phi \hat{\varphi}$$

therefore

$$\begin{aligned} \nabla \phi &= \left(-\frac{1}{r^2} e^{-Kr} - \frac{K}{r} e^{-Kr} \right) \hat{r} \\ &= - \left(\frac{1}{r^2} + \frac{K}{r} \right) e^{-Kr} \hat{r} \end{aligned}$$

3.2 laplacian

The laplacian in spherical coordinates is defined as

$$\nabla^2 \phi = \frac{1}{r^2} \partial_r (r^2 \partial_r \phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta \phi) + \frac{1}{r^2 \sin^2 \theta} \partial_\varphi^2 \phi$$

therefore

$$\begin{aligned} \nabla^2 \phi &= \frac{1}{r^2} \partial_r (- (1 + Kr) e^{-Kr}) \\ &= \frac{1}{r^2} (-K e^{-Kr} + (1 + Kr) K e^{-Kr}) \\ &= \frac{1}{r^2} (K^2 r e^{-Kr}) \\ &= \frac{K^2}{r} e^{-Kr} \end{aligned}$$

3.3 Gauss' law

Gauss' law for this subsection is

$$\iiint_V \nabla^2 \phi d\mathbf{x} = \int_{\partial V} \nabla \phi \cdot d\mathbf{a}$$

therefore, using the result from subsection 2

$$\iiint_V \nabla^2 \phi d\mathbf{x} = \int_{\partial V} - \left(\frac{1}{r^2} + \frac{K}{r} \right) e^{-Kr} \hat{\mathbf{r}} \cdot d\mathbf{a}$$

We know that $d\mathbf{a}$ points in the radial direction therefore $\hat{\mathbf{r}} \cdot d\mathbf{a} = da$ and thus

$$\begin{aligned} \iiint_V \nabla^2 \phi d\mathbf{x} &= \int_{\partial V} - \left(\frac{1}{r^2} + \frac{K}{r} \right) e^{-Kr} da \\ &= - \left(\frac{1}{R^2} + \frac{K}{R} \right) e^{-KR} \int_{\partial V} da \\ &= - \left(\frac{1}{R^2} + \frac{K}{R} \right) e^{-KR} \cdot 4\pi R^2 \\ &= -4\pi (1 + KR) e^{-KR} \end{aligned}$$

Comparing this to a direct calculation

$$\begin{aligned} \iiint_V \nabla^2 \phi d\mathbf{x} &= \iiint_V \frac{K^2}{r} e^{-Kr} d\mathbf{x} \\ &= \int_0^{2\pi} d\varphi \int_0^\pi \sin \theta d\theta \int_0^R r^2 \frac{K^2}{r} e^{-Kr} dr \\ &= 4\pi K^2 \int_0^R r e^{-Kr} dr = \left[U = r \quad U' = 1 \atop V' = e^{-Kr} \quad V = -\frac{1}{K} e^{-Kr} \right] \\ &= 4\pi K^2 \left(-\frac{r e^{-Kr}}{K} \Big|_0^R + \frac{1}{K} \int_0^R e^{-Kr} dr \right) \\ &= 4\pi K^2 \left(-\frac{r e^{-Kr}}{K} \Big|_0^R + \frac{1}{K} \left(-\frac{1}{K} e^{-Kr} \right) \Big|_0^R \right) \\ &= 4\pi K^2 \left(-\frac{R e^{-KR}}{K} + \frac{1}{K} \left(\frac{1}{K} - \frac{1}{K} e^{-KR} \right) \right) \\ &= 4\pi K^2 \left(-\frac{R e^{-KR}}{K} + \frac{1}{K^2} - \frac{1}{K^2} e^{-KR} \right) \\ &= 4\pi (-K R e^{-KR} + 1 - e^{-KR}) \\ &= 4\pi (1 - (1 + KR) e^{-KR}) \end{aligned}$$

We notice that the direct calculation gives an answer that is 4π off from the result of Gauss' law, which is likely a result of the fact our laplacian definition did not include the origin. Thus we'll add a correction for the origin:

$$\nabla^2 \phi = \frac{K^2}{r} e^{-Kr} - 4\pi \delta^3(\mathbf{r})$$

3.4 integral of laplacian

$$\iiint d^3x \nabla^2 \phi = \iiint d^3x \left(k^2 \frac{e^{-Kr}}{r} - 4\pi \delta(\mathbf{r}) \right) \quad (14)$$

but

$$\iiint d^3x k^2 \frac{e^{-Kr}}{r} = 4\pi K^2 \int_0^\infty dr r e^{-Kr} \quad (15)$$

and since

$$\int_0^\infty dr r e^{-Kr} = \frac{1}{K^2} \quad (16)$$

then

$$4\pi K^2 \int_0^\infty dr r e^{-Kr} = 4\pi \quad (17)$$

and the second part

$$\iiint d^3x 4\pi \delta(\mathbf{r}) = 4\pi \quad (18)$$

adding the two parts together

$$\iiint d^3x k^2 \frac{e^{-Kr}}{r} = 4\pi K^2 \int_0^\infty dr r e^{-Kr} = 4\pi - 4\pi = 0 \quad (19)$$